

Operators as Boundary Spectral Relations: Exact Magnetic-Graph Synthesis and Fixed-Boundary Cobordism Relaxation

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Abstract

We study a spectral encoding of a finite-dimensional linear operator in which the operator is not identified with one fitted eigenvector. Instead, the graph of a map $U : \mathcal{H}_- \rightarrow \mathcal{H}_+$ is required to equal the boundary trace of a common eigenspace of a weighted magnetic connection Laplacian. This formulation makes input substitution a consequence of linearity: fitting a basis at one common eigenvalue fits every superposition, provided that the boundary trace has no additional modes and its input projection is nonsingular. We prove an exact finite-network realization theorem. For any Hermitian boundary matrix, any fixed induced boundary magnetic graph, and any positive spectral parameter, a finite simple magnetic graph can be constructed whose Schur response is exactly the prescribed matrix while every intrinsic boundary edge remains unchanged. Applying the theorem to a canonical graph-of-map penalty realizes every complex linear map, including singular and non-normal maps. Charge conservation is characterized separately as equivariance of the boundary relation; it is a symmetry condition, not an existence condition. Boundary permutations are shown to be framing data and therefore part of operator identification.

We then describe a fixed-boundary relaxation implemented in Tessera. The method pins incoming and outgoing restrictions for a complete training basis, varies only bulk geometry and bulk witness amplitudes, and minimizes the full-complex connection-Laplacian eigenresidual at a common eigenvalue. Deterministic tests recover a generic two-dimensional unitary to relative error 2.51×10^{-16} with zero recorded boundary drift. A completed 77-case stress test verifies 43 cases, exhausts the optimization budget in 22, rejects eight preparations under the current normalization and isolated-eigenstate preconditions, and exposes four identifiability or common-eigenvalue failures. These are numerical solver outcomes, not a universality proof. The exact theorem is presently a magnetic-graph theorem. Proving that the construction can always be thickened into a nondegenerate triangulated cobordism without changing its boundary response remains the central geometric problem. Accordingly, the work establishes an exact operator-as-spectral-relation framework and a constructive graph realization, but does not claim a topological quantum field theory or a universal simplicial realization theorem.

Keywords. inverse spectral problem; magnetic graph Laplacian; Schur complement; cobordism; quantum operator; charge equivariance; fixed-boundary relaxation

1 Introduction

A cobordism $W : M_- \rightsquigarrow M_+$ is geometrically interpreted as a process between an incoming and an outgoing boundary. In a topological quantum field theory (TQFT), a functor assigns a linear map

to W , and gluing cobordisms corresponds to composition of maps [1]. The inverse question is less standard: can a prescribed linear map be represented by finite geometry, and can that representation be found by relaxation while the boundary data remain fixed?

The motivating numerical experiments use weighted simplicial complexes. A boundary state is represented in a selected spectral frame on a boundary component. Given an algebraic operator U , the expected outgoing state is computed first. Incoming and outgoing restrictions are then pinned, while bulk weights, connection phases, topology, and auxiliary interior amplitudes are adjusted to reduce a full-complex eigenresidual. Once one geometry has been fitted using a complete basis, held-out superpositions are evaluated with the same geometry.

There are three materially different claims that must not be conflated.

- (i) A single vector proportional to $\text{vec}(U)$ can be prepared as an eigenstate. This is state synthesis in a Choi-like coordinate system, not by itself an operational representation of U .
- (ii) Boundary traces of one common eigenspace can form the graph $\{(x, Ux) : x \in \mathcal{H}_-\}$. This is an operational representation: changing x changes the outgoing restriction to Ux by linearity.
- (iii) A scalar weighted magnetic network realizing that spectral relation can always be chosen to be the one-skeleton of a valid triangulated cobordism with the prescribed frozen boundary. This additional relative-realization claim is open.

This paper proves the second claim abstractly and gives an exact finite magnetic graph construction. It reports numerical evidence for the corresponding fixed-boundary relaxation, while isolating the unproved step needed for the strongest simplicial-cobordism interpretation.

1.1 Contributions

The principal contributions are as follows.

1. We define an operator represented by a cobordism as a *boundary spectral relation*, rather than as a distinguished bulk vector.
2. We prove that complete-basis witnesses at a common eigenvalue extend to every superposition, and state the rank condition required to exclude spurious boundary modes.
3. We give a canonical positive semidefinite matrix K_U whose kernel is exactly the graph of an arbitrary complex linear map U .
4. We prove constructively that every Hermitian matrix is the boundary Schur response of a finite simple scalar magnetic graph at any prescribed positive spectral parameter, without changing the induced boundary graph.
5. We prove that charge conservation is equivalent to invariance of the graph relation, and to commutation of K_U with total boundary charge.
6. We identify attachment permutations as boundary framings and give residual-, gap-, and conditioning-based certification criteria.
7. We document the exact semantics and current numerical limits of the Tessera relaxation experiments.

2 Relation to prior work

The functorial assignment of state spaces and linear maps to manifolds and cobordisms is axiomatized by Atiyah [1]. Our direction is inverse: the map is specified first, and a finite spectral network is synthesized to carry its graph. We do not establish the identity, gluing, monoidal, or diffeomorphism-invariance axioms required of a TQFT.

Encoding an operator as a vector is familiar from the Jamiolkowski–Choi correspondence [2, 3]. That correspondence is useful for preparing training targets, but a fitted vector is not automatically a device that accepts a new input. The graph-of-map formulation used here supplies the missing input/output relation.

Eliminating interior vertices by a Schur complement is Kron reduction [4]. Closely related inverse response questions have been studied for metric quantum graphs: Friedlander showed that arbitrary real symmetric matrices can occur as Dirichlet-to-Neumann maps for a quantum-graph Schrödinger operator [5]. Inverse eigenvalue questions for ordinary graph Laplacians remain constrained by graph structure [6]. Magnetic phases enlarge the realizable class from real symmetric to complex Hermitian matrices; magnetic and connection Laplacians are standard objects in spectral graph theory and data analysis [7, 8]. Our finite construction is explicit, preserves a preassigned induced boundary graph, and is tailored to a graph-of-map kernel.

The triangulated extension is related to, but not resolved by, bistellar equivalence of triangulations [9]. Bistellar or stellar moves control topology; they do not automatically preserve a prescribed magnetic Schur response.

3 The discrete spectral object

3.1 Scalar magnetic connection Laplacian

Let $G = (V, E)$ be a finite simple graph. Give each unoriented edge $\{u, v\}$ a weight $w_{uv} > 0$ and an oriented phase $a_{uv} = e^{i\theta_{uv}} \in U(1)$, with $a_{vu} = \overline{a_{uv}}$. The scalar magnetic connection Laplacian L_G acts on $z \in \mathbb{C}^V$ by

$$(L_G z)_u = \sum_{v: \{u, v\} \in E} w_{uv} (z_u - a_{uv} z_v). \quad (3.1)$$

It is Hermitian and positive semidefinite because

$$z^\dagger L_G z = \sum_{\{u, v\} \in E} w_{uv} |z_u - a_{uv} z_v|^2. \quad (3.2)$$

In the Tessera implementation, w_{uv} is the positive quantity derived from the squared edge length and θ_{uv} is the optimized real connection phase.

This distinction is important: the experiments analyzed here use the degree-zero $\mathbb{C}^*$ connection Laplacian in (3.1). They do not use a degree-one Hodge Laplacian, and they do not minimize a Regge action or a harmonic-period residual. The ordinary degree-zero discrete exterior calculus Laplacian is a separate operator in the codebase.

3.2 Boundary response

Choose a nonempty boundary vertex set $B \subseteq V$ and write $I = V \setminus B$. Relative to $\mathbb{C}^V = \mathbb{C}^B \oplus \mathbb{C}^I$, write

$$L_G = \begin{pmatrix} A & C \\ C^\dagger & D \end{pmatrix}. \quad (3.3)$$

For a real spectral parameter $\lambda \notin \text{spec}(D)$, define the boundary Schur response

$$S_G(\lambda) = A - \lambda I_B - C(D - \lambda I_I)^{-1} C^\dagger. \quad (3.4)$$

Definition 3.1 (Boundary spectral relation). The boundary spectral relation of (G, B) at λ is

$$\mathcal{R}_G(\lambda) = \{z|_B : z \in \ker(L_G - \lambda I_V)\} \subseteq \mathbb{C}^B. \quad (3.5)$$

Proposition 3.2 (Schur characterization). *If $\lambda \notin \text{spec}(D)$, then*

$$\mathcal{R}_G(\lambda) = \ker S_G(\lambda), \quad z_I = -(D - \lambda I_I)^{-1} C^\dagger z_B. \quad (3.6)$$

In particular, every admissible boundary trace has a unique interior lift.

Proof. The interior block of $(L_G - \lambda I)z = 0$ gives the second identity. Substitution into the boundary block gives $S_G(\lambda)z_B = 0$. The converse follows by reversing these steps. $\square$

4 An operator is the graph of a boundary relation

Let $\mathcal{H}_- \cong \mathbb{C}^d$ and $\mathcal{H}_+ \cong \mathbb{C}^m$ be incoming and outgoing state spaces, and let $U : \mathcal{H}_- \rightarrow \mathcal{H}_+$ be linear. Its graph is

$$\Gamma_U = \{x \oplus Ux : x \in \mathcal{H}_-\} \subseteq \mathcal{H}_- \oplus \mathcal{H}_+. \quad (4.1)$$

Physical boundary vertices need not be coordinate vectors of these spaces. Let $J_- : \mathcal{H}_- \hookrightarrow \mathbb{C}^{B_-}$ and $J_+ : \mathcal{H}_+ \hookrightarrow \mathbb{C}^{B_+}$ be fixed injective state charts on two disjoint boundary components. For isometric charts, the embedded graph is the range of

$$T_U x = J_- x \oplus J_+ Ux. \quad (4.2)$$

Definition 4.1 (Exact spectral realization). Let $B = B_- \sqcup B_+$. A framed magnetic graph (G, B, J_-, J_+) realizes U at λ if

$$\mathcal{R}_G(\lambda) = \text{ran } T_U. \quad (4.3)$$

Equivalently, after applying the inverse charts, the relation is Γ_U .

The equality in (4.3), rather than mere containment, is the identifiability requirement. It rules out an additional eigenvector with zero incoming trace or a second outgoing trace for the same input.

4.1 Why a complete basis generalizes

Theorem 4.2 (Basis-span transfer). *Let $b_1, \dots, b_d$ be a basis of $\mathcal{H}_-$. Suppose that for one graph G and one common eigenvalue λ , there are witnesses $z_1, \dots, z_d \in \ker(L_G - \lambda I)$ satisfying*

$$z_j|_{B_-} = J_- b_j, \quad z_j|_{B_+} = J_+ U b_j. \quad (4.4)$$

Then $\text{ran } T_U \subseteq \mathcal{R}_G(\lambda)$. Consequently every $x \in \mathcal{H}_-$ has a full-graph eigenvector whose incoming and outgoing restrictions are $J_- x$ and $J_+ Ux$, respectively. If additionally $\dim \mathcal{R}_G(\lambda) = d$, then G realizes U exactly.

Proof. Write $x = \sum_j \alpha_j b_j$ and set $z = \sum_j \alpha_j z_j$. Linearity gives $(L_G - \lambda I)z = 0$, and restriction gives $z|_B = T_U x$. Thus $\text{ran } T_U \subseteq \mathcal{R}_G(\lambda)$. Since T_U is injective, $\dim \text{ran } T_U = d$; equality follows under the stated dimension condition. $\square$

Remark 4.3 (The eigenvector is global). The output is not, in general, an eigenvector supported only on B_+ . It is the outgoing restriction of a full-graph eigenvector. The bulk amplitudes are the unique lift in [Proposition 3.2](#) when the interior block is nonsingular at λ .

Remark 4.4 (A common eigenvalue is essential). If witness z_j belongs to eigenvalue λ_j and the λ_j are not equal, a generic linear combination is not an eigenvector. Small individual residuals therefore do not establish operator transfer unless the eigenvalue spread is also controlled.

4.2 Recovering the represented matrix

Let $N \in \mathbb{C}^{(|B_-|+|B_+|) \times d}$ contain a basis of the boundary relation, expressed in the selected charts, and split it as

$$N = \begin{pmatrix} X \\ Y \end{pmatrix}. \quad (4.5)$$

The relation is the graph of a unique operator if and only if X is square and nonsingular; in that case

$$U = YX^{-1}. \quad (4.6)$$

This rank test is the correct operational test. A vectorized bulk state is not itself a map. If desired, one may vectorize the recovered matrix in (4.6) to obtain its Jamiolkowski–Choi vector or state.

5 A universal algebraic target

There is a canonical Hermitian matrix whose kernel is exactly Γ_U . Set

$$F_U = \begin{pmatrix} -U & I_m \end{pmatrix}, \quad K_U = F_U^\dagger F_U = \begin{pmatrix} U^\dagger U & -U^\dagger \\ -U & I_m \end{pmatrix}. \quad (5.1)$$

For $z = x \oplus y$,

$$z^\dagger K_U z = \|y - Ux\|^2, \quad \ker K_U = \Gamma_U. \quad (5.2)$$

The nonzero eigenvalues of K_U are the eigenvalues of $F_U F_U^\dagger = I_m + UU^\dagger$, so its nonzero spectral gap is at least one. If U is unitary, $K_U^2 = 2K_U$ and $K_U/2$ is the orthogonal projector onto $\Gamma_U^\perp$.

For embedded isometric charts, a coordinate-free target on the full boundary space is

$$P_U = I_B - T_U(T_U^\dagger T_U)^{-1} T_U^\dagger, \quad \ker P_U = \text{ran } T_U. \quad (5.3)$$

Thus universal operator encoding reduces to realizing a prescribed Hermitian boundary response with a prescribed kernel.

6 Charge symmetry and boundary framing

6.1 Charge is equivariance, not existence

Let Q_- and Q_+ be Hermitian charge operators on $\mathcal{H}_-$ and $\mathcal{H}_+$, and set $Q_\partial = Q_- \oplus Q_+$. The usual charge-conservation condition is

$$Q_+ U = U Q_-. \quad (6.1)$$

Theorem 6.1 (Equivalent charge criteria). *For any linear U , the following statements are equivalent:*

- (a) $Q_+ U = U Q_-$;

$$(b) \quad Q_{\partial}\Gamma_U \subseteq \Gamma_U;$$

$$(c) \quad [K_U, Q_{\partial}] = 0.$$

Proof. For $x \oplus Ux \in \Gamma_U$, invariance under Q_{∂} is precisely the condition $Q_+Ux = UQ_-x$ for every x . This proves (a) $\Leftrightarrow$ (b). The lower-left block of $[K_U, Q_{\partial}]$ vanishes exactly when $Q_+U = UQ_-$. Under this identity and its adjoint, $[U^{\dagger}U, Q_-] = 0$, so all remaining blocks vanish. Hence (a) $\Leftrightarrow$ (c). $\square$

The existence theorem below does not require (6.1). Charge conservation becomes necessary only when the realizing geometry is also required to respect the corresponding boundary symmetry. In particular, a charge-changing Pauli X map is a valid spectral relation even though it is not an intertwiner for the computational Z charge. Inside a degenerate sector on which $Q_- = Q_+ = qI$, every operator is automatically equivariant; the charge condition then imposes no additional restriction.

6.2 Attachment permutations are framing data

The matrices assigned to a boundary depend on ordered state charts. Under invertible changes of incoming and outgoing frame P_- and P_+ , the same abstract relation has coordinate matrix

$$U \mapsto P_+UP_-^{-1}. \quad (6.2)$$

Vertex permutations are special cases. Therefore the way an input complex is attached to the bulk is not an incidental implementation detail: it specifies the boundary framing in which the operator is read.

Trying every attachment permutation is a valid way to identify an unknown framing. Once selected, however, the same permutation must be held fixed for the complete training basis and every held-out input. Selecting a new permutation separately for each state would fit a state-dependent rule rather than one operator. Moreover, a successful permutation does not by itself resolve rank deficiency: the input block X in (4.6) must remain well conditioned.

7 Exact synthesis of arbitrary boundary response

We now show that the algebraic target in Equation (5.1) can always be realized by a finite scalar magnetic network. The result preserves every intrinsic edge of a preassigned boundary graph. New boundary-to-interior edges contribute to the full diagonal block, as they must, but do not alter the induced boundary subgraph.

Theorem 7.1 (Fixed-boundary magnetic response synthesis). *Let B be a finite nonempty vertex set, let G_0 be any weighted magnetic graph induced on B , let $R \in \mathbb{C}^{B \times B}$ be Hermitian, and let $c > 0$. There exists a finite simple weighted magnetic graph G with boundary B such that:*

- (i) *the subgraph induced by B is exactly G_0 , with all of its weights and phases unchanged;*
- (ii) *$c \notin \text{spec}(L_{II})$; and*
- (iii) *$S_G(c) = R$.*

If desired, G may be chosen connected.

Proof. Let L_0 denote the magnetic Laplacian contribution of the fixed boundary graph, embedded in the boundary block. We construct mutually nonadjacent interior vertices, so L_{II} will be diagonal.

For each unordered pair $i < j$, define the required off-diagonal correction

$$\delta_{ij} = R_{ij} - (L_0)_{ij}. \quad (7.1)$$

If $\delta_{ij} \neq 0$, add an interior vertex v_{ij} connected only to i and j . Give both edges a common weight $t \in (0, c/2)$ and choose their relative phase to be $\arg \delta_{ij}$. Since the diagonal entry of L_{II} at this vertex is $2t$, its off-diagonal contribution to the Schur response is

$$\frac{t^2}{c - 2t} e^{i \arg \delta_{ij}}. \quad (7.2)$$

For $a = |\delta_{ij}|$, the choice

$$t = \sqrt{a^2 + ca} - a \quad (7.3)$$

lies in $(0, c/2)$ and satisfies $t^2/(c - 2t) = a$. Hence the pair gadget produces exactly δ_{ij} and its conjugate in the two off-diagonal positions. Its contribution to each incident diagonal is

$$g_c(t) = t + \frac{t^2}{c - 2t} > 0. \quad (7.4)$$

After all pair gadgets have been inserted, only the diagonal entries may be incorrect. A leaf joined only to boundary vertex i with edge weight $p > 0$, $p \neq c$, contributes

$$f_c(p) = p - \frac{p^2}{p - c} = \frac{pc}{c - p} \quad (7.5)$$

to the i th response diagonal. The range of f_c on $(0, c)$ is $(0, \infty)$, and its range on (c, ∞) is $(-\infty, -c)$. Thus every real diagonal correction η is a sum of at most two leaf responses. If $\eta > 0$ or $\eta < -c$, one leaf suffices, with

$$p = \frac{c\eta}{c + \eta} > 0. \quad (7.6)$$

If $-c \leq \eta < 0$, write $\eta = 2c + (\eta - 2c)$; the first term lies in $(0, \infty)$ and the second in $(-\infty, -c)$, so two leaves suffice. No leaf is needed for $\eta = 0$.

The interior diagonal entries are either $2t < c$ or $p \neq c$, so $c \notin \text{spec}(L_{II})$. The construction uses distinct interior vertices and no parallel edges, hence G is simple, and no edge of G_0 was changed.

If the resulting graph is disconnected, choose a spanning tree on its boundary components. For every tree edge, insert two pair gadgets having the same weight and opposite relative phases. Their off-diagonal responses cancel, while they connect the components. Compensate their diagonal contributions with the leaf construction above. This preserves $S_G(c) = R$ and yields a connected graph. $\square$

Corollary 7.2 (Universal graph realization of a linear map). *For every complex linear map $U : \mathbb{C}^d \rightarrow \mathbb{C}^m$, every fixed induced boundary magnetic graph on $d + m$ coordinate vertices, and every $c > 0$, there is a finite simple magnetic graph whose boundary spectral relation at c is exactly Γ_U . The same conclusion holds in fixed injective boundary charts by using the target P_U in Equation (5.3).*

Proof. Apply Theorem 7.1 with $R = K_U$ in direct coordinates, or with $R = P_U$ in embedded charts. By Proposition 3.2, the boundary relation is the kernel of the response, which is the desired graph. $\square$

The construction is an existence proof, not an efficient optimizer. It uses at most one degree-two interior vertex for each nonzero off-diagonal correction and at most two leaves per boundary vertex, before optional connectivity gadgets. It applies to rectangular, singular, non-normal, charge-changing, and poorly conditioned operators.

8 What remains to obtain a simplicial-cobordism theorem

Every finite graph is a one-dimensional simplicial complex. That observation does not prove the geometric claim of primary interest. A one-dimensional complex with branching interior vertices is generally not a cobordism between the chosen boundary components, and thickening it into a higher-dimensional triangulated manifold introduces additional edges and diagonal contributions that may change (3.4).

Open problem 8.1 (Relative simplicial realization). Given the magnetic graph produced by [Theorem 7.1](#), construct a nondegenerate triangulated cobordism W with the prescribed frozen boundary such that the degree-zero connection Laplacian of W has the same boundary Schur response at c , possibly after adding response-neutral interior gadgets.

A proof would upgrade [Corollary 7.2](#) from finite discrete geometry to the specific class of simplicial cobordisms used by the numerical machinery. It must address four constraints simultaneously:

- (1) manifold links at every interior and boundary simplex;
- (2) nondegenerate simplex geometry and compatible shared edge lengths;
- (3) a fixed induced boundary triangulation and fixed boundary phases; and
- (4) exact preservation, or controllably accurate approximation, of the Schur response at the selected spectral parameter.

Pachner connectivity alone does not solve the problem, because a sequence of bistellar moves need not preserve spectral response. A possible alternative for unitary maps is a rank- d non-Abelian connection on a cylinder whose holonomy is U . That approach makes the operator geometrically immediate but changes the scalar $U(1)$ machinery used in the present experiments; arbitrary nonunitary maps would additionally require dilation or a nonunitary bundle connection.

9 Fixed-boundary relaxation

The constructive proof above was derived to clarify what the numerical solver is attempting to discover. The present Tessera experiment uses two disjoint triangulated boundary components and a shared bulk. Its reference fixture places a three-edge cycle on each boundary. At unit weights and zero phases, the cycle is K_3 , whose connection-Laplacian spectrum is $\{0, 3, 3\}$. The two nonconstant Fourier modes provide a two-dimensional neutral state chart; an optional constant mode has eigenvalue zero.

For a chosen basis $b_1, \dots, b_d$, the expected outputs Ub_j are computed algebraically. Their boundary representations are fixed as in [Equation \(4.4\)](#). The solver then varies bulk edge weights, bulk connection phases, bulk witness amplitudes, and—when enabled—bulk stellar growth. Intrinsic boundary edge weights and phases remain bitwise fixed, and every witness retains its exact incoming and outgoing restriction.

If $\hat{z}_j$ denotes a full-complex witness and $\bar{\lambda}$ the common Rayleigh target, the coupled objective is

$$\mathcal{J}(W, \{\hat{z}_j\}) = \sum_{j=1}^d \left\| L_W \hat{z}_j - \bar{\lambda} \hat{z}_j \right\|^2. \quad (9.1)$$

Optimization uses bounded positive bulk weights, bounded real phases, Levenberg–Marquardt restarts, and optional boundary-preserving stellar growth. No Regge-stationarity term, curvature action, period objective, or explicit charge penalty appears in (9.1).

The protocol is therefore:

1. prepare each boundary chart in isolation and record the intended degenerate spectral sector;
2. select U and compute Ub_j algebraically for a complete basis;
3. pin all incoming and outgoing restrictions and freeze intrinsic boundary geometry;
4. relax only the shared bulk variables against the full- W residual;
5. certify common eigenvalue, boundary invariance, graph dimension, input rank, operator readout, and held-out superpositions; and
6. when attachment framing is unknown, enumerate its permutations, but retain one selected framing for all training and test states.

This is a supervised inverse spectral construction. The target outputs are used during fitting. After fitting, basis-span transfer is exact in principle, but only if the common-eigenvalue and identifiability hypotheses of [Theorem 4.2](#) have been established.

10 Numerical evidence

All values in this section are solver records, not mathematical assumptions. The boundary-drift and restriction metrics are evaluated independently of the optimization target. Every eigenresidual field, including a held-out full- W residual, is a squared norm. Boundary and operator errors are ordinary norms unless labeled otherwise. Therefore a recorded objective r certifies a residual norm no larger than $\sqrt{r}$ for each witness.

10.1 Historical direct-state mode

The earlier experiment pinned one vectorized operator target. It established that the relaxation can synthesize both charge-preserving and charge-changing states, but it did not establish input substitution. Representative outcomes are shown in [Table 1](#).

Table 1: Historical direct-state synthesis. Residual is the recorded squared objective.

Target	$\ [U, Q]\ _F$	residual	growth	boundary drift
phase gate	0	7.43×10^{-28}	3	0
Pauli X	1.41421	2.69×10^{-27}	2	0

The corresponding residual-norm upper bounds are approximately 2.73×10^{-14} and 5.19×10^{-14} . Pauli X succeeds despite its nonzero charge commutator, directly falsifying charge conservation as a necessary condition for unconstrained spectral state synthesis.

10.2 Coupled graph-of-map fixture

The current coupled experiment fits a complete two-dimensional basis at a common eigenvalue on one shared geometry. For a fixed generic 2×2 unitary, the default run produced the metrics in [Table 2](#).

Table 2: Deterministic coupled operator-transfer fixture.

Metric	Recorded value
full- W squared residual	7.36×10^{-17}
stellar growth steps	7
intrinsic boundary drift	0
pinned restriction error	0
maximum isolated-boundary squared residual	1.22×10^{-30}
recovered-operator relative error	2.51×10^{-16}
maximum held-out full- W squared residual	1.36×10^{-16}
maximum held-out boundary error	1.84×10^{-16}
common-eigenvalue spread	2.88×10^{-10}

The operator readout and held-out boundary restrictions are at floating-point precision. The full- W objective corresponds to a residual-norm bound of approximately 8.58×10^{-9} , so it should not be described as a machine-precision eigenpair. The evidence supports accurate transfer in this fixture, not exact numerical stationarity.

10.3 Completed stress test

A completed, confirmed smoke campaign contains 77 cases spanning named gates, random unitary, normal, non-normal, scaled, singular, charge-preserving, charge-permuting, and charge-mixing matrices; and a sequence of increasingly conditioned input frames. The status counts are in Table 3.

Table 3: Outcomes of the completed 77-case stress test. “Budget” means the configured optimizer budget ended before certification; it is not a proof of nonexistence.

Status	Count	Fraction
verified	43	55.8%
optimization budget exhausted	22	28.6%
boundary target rejected	8	10.4%
operator readout unstable	3	3.9%
fitted span not at one common eigenvalue	1	1.3%
total	77	100%

Five rejections were zero training outputs, which are valid for singular maps but incompatible with the current witness normalization. One input and two outputs failed the isolated-boundary eigenstate precondition. No case changed a frozen boundary edge or a pinned boundary restriction. Among verified cases, the largest recorded squared residual was 9.89×10^{-17} , the largest relative operator error was 2.64×10^{-10} , the largest held-out relative operator error was 3.89×10^{-16} , the largest held-out full squared residual was 9.83×10^{-16} , and the largest common-eigenvalue spread was 1.77×10^{-9} . Verified operator condition numbers ranged from 1 to 5.45×10^{16} and operator norms from 0.141 to 10^3 ; these ranges do not imply uniform coverage within the interval.

The largest verified training readout error came from a nearly collinear input frame. The conditioning sequence in Table 4 shows the expected loss of identifiability.

A separate deterministic set of nine named unitaries verified all nine, with recorded squared residuals between 6.26×10^{-17} and 9.11×10^{-17} , zero boundary violations, and at most seven growth

Table 4: Input-frame conditioning and operator readout.

input condition number	relative readout error	status
2×10^2	3.95×10^{-15}	verified
2×10^4	2.67×10^{-12}	verified
2×10^6	2.64×10^{-10}	verified
2×10^8	1.17×10^{-8}	unstable
2×10^{10}	9.63×10^{-7}	unstable
2×10^{12}	1.88×10^{-4}	unstable

steps. Two larger shuffled all-attachment campaigns were still running at the freeze date of this draft. No conclusion in this manuscript depends on their partial records.

These results support three limited conclusions. First, the coupled solver can recover one operator from a complete basis and transfer held-out states on the same frozen geometry. Second, neither charge conservation nor unitarity is a necessary numerical precondition. Third, conditioning and common-eigenvalue certification are genuine failure modes. The results do not establish that the optimizer succeeds for every map or that every successful network is a valid simplicial cobordism.

11 Certification and stability

Let L be Hermitian and let E_λ be an eigenspace separated from the rest of the spectrum by

$$\delta = \text{dist}(\lambda, \text{spec}(L) \setminus \{\lambda\}) > 0. \quad (11.1)$$

For any unit vector v ,

$$\text{dist}(v, E_\lambda) \leq \frac{\|(L - \lambda I)v\|}{\delta}. \quad (11.2)$$

Thus a small eigenresidual has geometric meaning only relative to a spectral gap. Subspace versions are instances of Davis–Kahan perturbation bounds [10]; residual-based Hermitian bounds are discussed, for example, by Stewart [11].

Operator readout has a separate conditioning factor. Suppose an exact boundary frame has blocks X and $Y = UX$, whereas the computed frame has $\hat{X} = X + E$ and $\hat{Y} = UX + F$. Then

$$\hat{U} - U = (F - UE)(X + E)^{-1}. \quad (11.3)$$

If $\|X^{-1}E\| < 1$, the norm satisfies

$$\|\hat{U} - U\| \leq \frac{(\|F\| + \|U\| \|E\|) \|X^{-1}\|}{1 - \|X^{-1}E\|}. \quad (11.4)$$

This explains why small witness residuals can coexist with an unstable matrix readout when training inputs are nearly dependent.

A defensible numerical certificate should report at least:

- (1) full-complex residual norms and whether stored metrics are squared;
- (2) the common-eigenvalue spread across basis witnesses;
- (3) the spectral gap separating the intended cluster;

- (4) dimension of the boundary trace and singular values of its input projection;
- (5) boundary geometry drift and pinned restriction error;
- (6) relative operator readout error in a declared framing;
- (7) held-out linear-combination errors without re-optimizing geometry or attachment; and
- (8) optimizer budget, restart count, and topology growth.

Permutation enumeration should additionally report both the best and the distribution over framings. A best-case result selected after observing the test input is not an operator certificate.

12 Composition and transition amplitudes

Graphs of maps compose as linear relations:

$$\Gamma_V \circ \Gamma_U = \{(x, z) : \exists y, y = Ux, z = Vy\} = \Gamma_{VU}. \quad (12.1)$$

This identity is the algebraic core of a cobordism composition law. It does not yet prove that gluing two triangulated realizations and eliminating the interface produces the required spectral relation without spurious modes. That statement requires an additional gluing theorem controlling the interface Schur complement and eigenvalue multiplicity.

When a framed spectral cobordism W realizes a unique operator, define

$$\mathfrak{Z}_\lambda(W) = U \quad \text{when} \quad \mathcal{R}_W(\lambda) = \Gamma_U. \quad (12.2)$$

For an incoming state $|\psi_B\rangle$ and outgoing test state $|\psi_A\rangle$, the induced scalar transition amplitude is then

$$Z(W_{AB}; \psi_A, \psi_B) = \langle \psi_A | \mathfrak{Z}_\lambda(W_{AB}) | \psi_B \rangle = \langle \psi_A | U | \psi_B \rangle. \quad (12.3)$$

In oriented cobordism notation one may write

$$W_{AB} = \text{geo}(U), \quad \partial W_{AB} = \overline{\text{geo}(\psi_B)} \sqcup \text{geo}(\psi_A), \quad (12.4)$$

where $\sqcup$ denotes disjoint union and the bar reverses the incoming orientation. Equations (12.3)–(12.4) are valid as notation for the extracted spectral map. They are not presently a derivation of a path integral or a proof that $\mathfrak{Z}_\lambda$ is a TQFT functor. Such a claim would require, at minimum, identity, gluing, monoidality, and invariance under the chosen equivalence of cobordisms.

13 Discussion

The graph-of-map formulation settles the basic semantics of input substitution. The learned object is neither an isolated boundary eigenvector nor the bulk with its boundary literally subtracted. It is the boundary relation obtained after the bulk has been eliminated. The bulk acts as a realization of a Schur response, and the operator is recovered from the kernel of that response.

This perspective also resolves the proposed Choi interpretation. A Choi vector stores all matrix entries and can serve as a training target, but promotion to an operator requires a declared tensor factorization and reshaping convention. By contrast, Γ_U already has operational input and output projections. The two descriptions are algebraically interconvertible after the frame has been fixed, but they answer different experimental questions.

The exact response-synthesis theorem is nontrivial and broad: a scalar positive-weight magnetic network can realize the graph of every finite complex linear map while preserving arbitrary intrinsic boundary geometry. It also shows that charge conservation cannot be the general existence criterion. Charge instead determines when the relation and its penalty respect a selected symmetry. This distinction matters for both theorem statements and numerical test design.

The scientific limitation is equally clear. The current universal theorem is a theorem about finite magnetic graphs, which are discrete geometries but need not be manifold cobordisms. The relaxation operates in a narrower class of triangulated complexes and has finite budgets. Numerical success across many operators cannot replace the relative simplicial realization theorem, and budget exhaustion cannot refute it. The most valuable next mathematical step is therefore a response-preserving thickening construction or a proof that such a construction is obstructed for some boundary data.

14 Conclusion

A finite-dimensional operator can be represented intrinsically as the graph of a boundary spectral relation. Complete-basis fitting at a common eigenvalue then supports every new input by linearity, provided the boundary trace has the correct dimension and a nonsingular input projection. Every such relation has a canonical Hermitian penalty, and every Hermitian boundary response admits an explicit realization by a finite simple scalar magnetic graph with unchanged intrinsic boundary geometry. Charge conservation is exactly equivariance of this relation; attachment permutations select its frame.

The Tessera experiments implement a fixed-boundary inverse spectral relaxation consistent with these semantics and demonstrate high-accuracy operator readout and held-out transfer in verified cases. They also expose optimization, common-eigenvalue, and conditioning failures. Establishing universal realization within nondegenerate triangulated cobordisms, and proving a gluing law sufficient for a functorial amplitude assignment, remain open. These boundaries separate the proven mathematical result from the proposed geometric generalization.

A Reproducibility protocol

The reference implementation is the open-source Tessera repository [12]. The principal executable examples are:

- `examples/cobordism/geometric_operators.py`; and
- `examples/cobordism/geometric_operator_sweep.py`.

They are backed by the coupled solver in `MultiCobordism` and `EigenstateSynthesis`. After building the Python bindings according to the repository instructions, the deterministic fixture and the completed 77-case record are reproduced by the following invocations:

```
python examples/cobordism/geometric_operators.py
python examples/cobordism/geometric_operator_sweep.py \
--profile smoke --jobs 4 --confirmation-attempts 2 \
--output /tmp/cobordism/geometric_operator_sweep_smoke_confirmed.json
```

The larger exhaustive campaign enumerates all six attachment permutations and uses independently seeded operator generation, schedule shuffling, input states, and optimizer generation. It checkpoints every ten cases and retains every status rather than discarding failures. Reproduction should record the repository revision, compiler and linear-algebra library, floating-point precision, all random seeds, and the complete JSON result. Confirmatory runs must preserve the frozen boundary and framing while changing optimizer seeds.

B Claim ledger

Table 5: Epistemic status of the principal claims.

Claim	Status	Location
A complete basis at one common eigenvalue transfers every superposition.	proved	Theorem 4.2
K_U has kernel Γ_U for every complex linear map.	proved	Equation (5.2)
Every Hermitian boundary response is realizable by a finite simple magnetic graph with fixed induced boundary graph.	proved	Theorem 7.1
Charge conservation is invariance of Γ_U and commutation of K_U with boundary charge.	proved	Theorem 6.1
The current relaxation can recover and transfer multiple operator families.	numerical evidence	Section 10
Every target response admits a valid nondegenerate triangulated-cobordism realization with the same frozen boundary.	open	Open problem 8.1
Geometric gluing realizes operator composition without spurious modes.	open	Equation (12.1)
The construction defines a TQFT or path integral.	not claimed	Equation (12.3)
Charge conservation is necessary for unconstrained operator realization.	false	Theorem 6.1 and Table 1
A fitted Choi-like vector alone is an operator accepting new inputs.	false without promotion	Equation (4.6)

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